Colouring minimal Cayley graphs: the road from Babai's conjectures to their resolution

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Abstract

In 1978 Babai conjectured that minimal Cayley graphs have a bounded chromatic number. In 1995 he changed his mind and conjectured the opposite.

1 AI/Data usage

"I want a circular graph in latex with 3 vertices connected in a circle by 3 edges, two of them are red, one of them is blue"

2 Introduction

3 Preliminaries

3.1 Standard notation for paper

Definition 3.1.1 (Chromatic number)**.** [1] The smallest number of colors required to color the vertices of any graph G, so that any adjacent vertices are never the same color, is called the chromatic number of G. Denoted as $\chi(G)$.

Definition 3.1.2 (Circuit)**.** [1] A *circuit* is a closed walk that can't traverse the same edges twice.

Definition 3.1.3 (Cycle)**.** [1] A *cycle* is a circuit that can't traverse the same vertices twice.

3.2 Cayley graphs with small examples

Definition 3.2.1 (Cayley Graphs)**.** Let G be a group, and let $S \subseteq G$ be a generating set, construct the graph $C(G, S)$ as follows:

- every $g \in G$ is a vertex,
- for every ordered pair (g, s), where $s \cdot g_1 = g_2$ with $s \in S$ and $g_1, g_2 \in G$, set an edge between g_1 and g_2, disallowing identical edges.

The resulting object, $C(G, S)$, is then called a Cayley Graph.

3.3 Minimal generating sets

3.4 Hypergraph coloring

4 Babai's question, 1978-1995

4.1 Minimal Cayley graphs are no-lonely-color

Definition 4.1.1 (Quasi-irreducible generating set)**.** Let G be a group, and let $S \subseteq G$ be a generating set. We call S *quasi-irreducible* if the group generated by S equals G, e.g.

$$\langle S \rangle = G,$$

and S can be well-ordered such that no element of S is contained in the group generated by the previous elements.

Example. Let $S = \{2, 3, 4\}$, then $3 - 2 = 1$. Since $\langle 1 \rangle = \mathbb{Z}$, also $\langle S \rangle = \mathbb{Z}$. Now observe that the the ordening $\{2, 4, 3\}$ fails since $4 \in \langle 2 \rangle$, but $\{4, 2, 3\}$ works since $2 \notin \langle 4 \rangle$ and $3 \notin \langle 4, 2 \rangle$. Hence S is a quasi-irreducible generating set of $\mathbb{Z}$.

Counterexample. Let $S = \{1, 2, 3\}$. Notice that

$$\{3, 2, 1\}, \text{ then } 2 \in \langle 1, 3 \rangle,$$

$$\{3, 1, 2\}, \text{ then } 1 \in \langle 2, 3 \rangle,$$

$$\{2, 3, 1\}, \text{ then } 1 \in \langle 2, 3 \rangle,$$

$$\{2, 1, 3\}, \text{ then } 3 \in \langle 1, 2 \rangle,$$

$$\{1, 3, 2\}, \text{ then } 3 \in \langle 1 \rangle,$$

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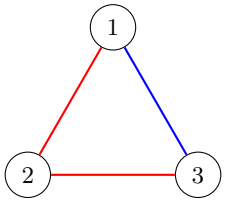
hence S is not a quasi-irreducible generating set of $\mathbb{Z}$.

Definition 4.1.2 (No-pied-circuit graph)**.** A graph G is a *no-pied-circuit* if there exists an edge-coloring such that

- every vertex's edges can only have at most two of the same color,
- each circuit contains at least two edges of the same color.

Example. figure out example

Counterexample.



Observe that every circuit only contains the color blue once, hence this graph is *not* a no-pied-circuit.

Definition 4.1.3 (No-lonely-color graphs)**.** A graph G is no-lonely-colored if there exists an edge-coloring such that

- every vertex's edges can only have at most two of the same color,
- every circuit contains no color exactly once.

Example.

Counterexample.

figure out counterexample

Proposition 4.2. *The definitions of no pied circuit graphs and no-lonely-color graphs can interchange circuit with cycle and nothing changes.*

make rigorous

Proof. □

Proposition 4.3. *no-lonely-color graph implies no-pied-circuit graph*

write proof

Proof. □

make rigorous

write proof

No pied circuit graphs arise from quasi-irreducible generating sets. Also every no lonely colored graph is a no pied circuit graph. Babai then conjectured that pied-circuits have a bounded chromatic number, which would mean every no lonely colored graph would also have bounded chromatic number.

4.4 Forbidden subgraphs

4.5 Babai's 1978 conjectures

Conjecture 1. *The chromatic number of any no-pied-circuit graph is bounded.*

Conjecture 2.

Both conjecture 2.2 and 3.5 from Babai's 1978 paper

4.6 The 1995 counter-conjecture

5 Evidence in both directions

5.1 Abelian groups are 3-colorable

5.2 More positive results

5.3 Negative results

6 Necessary tools

7 No-lonely-color graphs of large chromatic number

7.1 Tranquil hypergraphs

7.2 Tranquility of C_d

7.3 The construction

7.4 The proof

8 Minimal Cayley graphs of large chromatic number

9 Open questions

10 Appendix

References

[1] Douglas B. West. *Introduction to Graph Theory.* Prentice Hall, 2nd edition, 2001.